

Servo Arm Control System

User Manual for Arduino & Python GUI

1. Product Overview

This software controls a 5-servo robotic arm using joysticks or serial commands.

Compatible with Arduino UNO and similar boards.

2. Safety Warning

Power Supply:

- After uploading code, switch from USB to external power supply
- Use 5V 1A or higher power adapter
- This ensures stable operation for board and servos

Overload Protection:

- Do not apply excessive force when grasping objects
- If servo stalls, current increases sharply
- Overcurrent will trigger board protection and restart

3. Hardware Connection

Joystick 1: X=A0, Y=A1, Button=D2

Joystick 2: X=A3, Y=A2, Button=D4

4. Arduino Setup

1. Open servo_control.ino in Arduino IDE
2. Select your board (Arduino UNO/Nano)
3. Select correct port
4. Upload the sketch
5. Open Serial Monitor (9600 baud)

5. Serial Commands

Command Format:

0 90 - Set Servo0 to 90 degrees

1 45 - Set Servo1 to 45 degrees

2 180 - Set Servo2 to 180 degrees

3 90 - Set Servo3 to 90 degrees

4 60 - Set Servo4 to 60 degrees

90 - Set ALL servos to 90 degrees

90 90 90 90 90 - Batch set all servos

Example:

> 0 90

S0:90